Cubic Gauss Sums: Normalization, Cube-Lattice Errors, Distance-Weighted Perron Localization and Gram Margins

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Abstract

We study the analytic interfaces in the assigned manuscript on an unconditional first moment for cubic Gauss sums. We preserve its Eisenstein primary-generator, trace, Fourier-measure and complex-phase conventions. Full Poisson summation replaces a cube-lattice Riemann error by an explicit omitted-origin term and a rapidly decreasing remainder. Distance-sensitive summation of the source's height-localized cell bounds reduces the absolute-error loss by a factor of the number of norm cells, without assuming that the arithmetic error scales with cell mass. We also give the sharp universal exponent margin for the source's single-short-factor multiplicity strategy. These are local refinements, not new cancellation theorems. An unresolved factor-of-two compatibility issue with the public rational-prime smoothed theorem is displayed explicitly. Arithmetic sieves, exceptional structured moments, metaplectic inputs and the global prime asymptotic are not independently certified here. Contribution assessment: C3/5, notable proved local error and localization improvements with explicit arithmetic and normalization boundaries.

1 Scope, source architecture and status

This is the M049 report, family 23, applicability L3. Applicability is not a proof-validity or contribution score. The source is the September 25, 2026 manuscript *An unconditional first moment for cubic Gauss sums* [1]. We read its eight TeX files in full, including the proof-map figure, and its README: 4143 lines in nine files. This is a literal source read, not verification of every imported theorem or a formal proof audit. Source hashes, actual display intervals and private unread complements are recorded in the separate read-scope ledger. Public references were refreshed on October 8, 2026 using public titles and URLs only. Confidential originals remained local.

The proof architecture and our boundaries are as follows. "Source" below means a statement made or derived in that manuscript, not a theorem certified by this report.

Source file	Role in the claimed proof	Treatment here
`paper.tex`	All-primary first moment, rational pairing, bibliography	Normalization checked; global constant compatibility unresolved.
`background-type1`	Hecke inputs, cubic sieve extensions, Voronoi, metaplectic mean square, Type I	Imported analytic/arithmetic inputs; cube completion and residue conventions retained.
`dual`	Two exceptional prime-convolution moments; full-sum duality, large values and Gram contradiction	Gram exponent choice sharpened; moment theorem not independently certified.
`dispersion`	Corrected complex square, exact diagonal, Poisson, cube frequencies and mixed term	Fourier conventions checked; cube discretization sharpened locally.
`decomposition`	Independent prime tuples and stopping-bin coefficients, roughness, sparse terms, SW	Architecture read; source arithmetic uniformity remains an obligation.
`perron`	Low/high height estimates, two norm-cell partitions, sharp truncation	Conditional distance-weighted cell improvement proved.
`angular`	Every fixed integer angular mode, twisted SW and Type I	Fixed-mode scope retained; no growing-mode or missing power-moment claim.

The source's essential restrictions matter. For $AB \asymp X$, $B \le X^{1/2}$, the range $B > X^{.40}$ requires bounded-length independently smooth prime convolutions with prime factors at least a fixed power of X; an extra coupled product cutoff is not allowed. For $B \le X^{.40}$ arbitrary divisor-bounded squarefree coefficients and sharp individual norm cells are permitted by its stated dispersion input. Low-height corrected dispersion additionally uses roughness and a sequence Siegel–Walfisz condition. The exact stopping-bin identity preserves independent coefficients by averaging selections with the reciprocal binomial multiplicity; it does not justify arbitrary correlated coefficients. The angular appendix fixes its integer mode before all limits.

2 Normalizations and the public compatibility issue

Set $\omega = e^{2\pi i/3}$, $\mathcal{O} = \mathbb{Z}[\omega]$, $K = \mathbb{Q}(\omega)$, $\lambda = 1 + 2\omega = \sqrt{-3}$, $\mathrm{N}\, z = |z|^2$, and $\mathrm{Tr}\, z = z + \bar{z}$. Put $e(t) = e^{2\pi i t}$ for real t. Ideals prime to 3 are represented by their unique primary generators $b \equiv 1 \pmod 3$. For the cubic residue character χ_b, extended by zero on nonunits, the source uses

$$G(b) = \frac{1}{\sqrt{\mathrm{N}\, b}} \sum_{v \bmod b} \chi_b(v) e(\mathrm{Tr}(v/b)), \qquad c_* = \frac{(2\pi)^{2/3}}{3\Gamma(2/3)}. \tag{1}$$

Its conventions give $|G(b)| = \mu^2(b)$ and, on primary arguments, $G(ab) = G(a)G(b)\overline{\chi_b(a)}$, with both sides zero for overlapping squarefree factors. These arithmetic identities are retained as source inputs.

2.1 Pairing and sharp-to-smooth conversion

For a rational split prime $p = \pi\bar{\pi}$, both primary generators are counted. Cubic reciprocity gives $\chi_\pi(\bar{\pi}) = \chi_{\bar{\pi}}(\pi)$; conjugation makes the two values conjugate, hence both are 1. Representing residues by rational integers, the additive coefficient is $\bar{\pi} \bmod \pi$. Its cubic character is 1, so rescaling it does not alter the rational cubic Gauss sum. The two multiplicative characters on $\mathbb{F}_p$ are conjugate, and

$\chi(-1) = 1$. Counting the three cube preimages of nonzero cubes therefore gives

$$G(\pi) + G(\bar{\pi}) = \frac{S_p}{\sqrt{p}}, \qquad S_p = \sum_{x \bmod p} e(x^3/p). \tag{2}$$

Inert primes of norm at most X contribute $O(\sqrt{X})$ by $|G| \le 1$. Thus the source's all-primary coefficient $(6/5)c_*$ implies a rational sharp coefficient $(3/5)c_*$ for $S_p/(2\sqrt{p})$.

Proposition 2.1 (Conversion of a sharp coefficient)**.** *Let complex numbers u_p satisfy $A(x) = \sum_{p \le x} u_p = Cx^{5/6}/\log x + o(x^{5/6}/\log x)$. For each $W \in C_c^1((0,\infty))$,*

$$\sum_p u_p W(p/X) = \frac{5}{6} C \, \frac{X^{5/6}}{\log X} \int_0^\infty W(y) y^{-1/6} \, dy + o_W(X^{5/6}/\log X).$$

Proof. Stieltjes integration by parts gives $-\int A(Xy) W'(y) \, dy$. On the fixed compact support, the asymptotic for A is uniform after scaling, and $\log(Xy)/\log X \to 1$ uniformly. Finally $-\int y^{5/6} W'(y) \, dy = (5/6) \int y^{-1/6} W(y) \, dy$. No positivity of u_p is required. □

Consequently the source's rational smoothed coefficient would be $c_*/2$. But the primary public Theorem 1.1 of Dunn–Radziwiłł v3 explicitly gives c_* under GRH for the same rational summand and smoothing; the journal abstract agrees [2, 3]. Their introduction also displays an all-primary sharp coefficient $(6/5)c_*$. The source acknowledges the rational/all-primary discrepancy in `paper.tex`, lines 182–192. The conflict is therefore not confined to a conjecture's printed constant: it includes the public smoothed theorem. We leave normalization compatibility unresolved and do not advertise a settled unconditional Patterson asymptotic. Changing c_* in only one model would also change its mixed and square terms; that is not a repair established here.

2.2 Trace Fourier measure

Let $dm(z) = 2 \, dx \, dy/\sqrt{3}$ and define, for $f \in \mathcal{S}(\mathbb{C})$,

$$\mathcal{F}_m f(y) = \widehat{f}(y) = \int_{\mathbb{C}} f(z) e(-\operatorname{Tr}(zy)) \, dm(z).$$

This gives $\mathcal{O}$ covolume one. Its trace-dual lattice is $\mathcal{O}^* = \lambda^{-1}\mathcal{O}$, of m-covolume $1/3$, and the primary coset $1 + 3\mathcal{O}$ has index 9.

Lemma 2.2 (Unitary versus lattice normalization)**.** *On Schwartz functions, extending by continuity to $L^2(dm)$,*

$$\mathcal{F}_m^2 f(x) = \tfrac{1}{3} f(-x), \qquad \|\mathcal{F}_m f\|_{L^2(dm)}^2 = \tfrac{1}{3} \|f\|_{L^2(dm)}^2 .$$

Thus $\sqrt{3}\mathcal{F}_m$ is unitary, whereas Poisson summation in this lattice normalization is $\sum_{j \in \mathcal{O}} f(j) = \sum_{k \in \mathcal{O}^*} \widehat{f}(k)$.

Proof. Writing $z = (x_1, x_2)$, $y = (y_1, y_2)$ gives $\operatorname{Tr}(zy) = 2(x_1 y_1 - x_2 y_2)$. Relative to the usual Euclidean transform, $\mathcal{F}_m f(y) = (2/\sqrt{3}) \mathcal{F}_{\mathrm{E}} f(2y_1, -2y_2)$. The linear substitution has absolute determinant 4. Applying it twice proves the inversion factor $(2/\sqrt{3})^2/4 = 1/3$; Euclidean Plancherel gives the same factor for the squared norm. The trace pairing on the basis $1, \omega$ has matrix $\left(\begin{smallmatrix} 2 & -1 \\ -1 & -1 \end{smallmatrix}\right)$, of determinant -3, verifying the dual lattice and covolume. Euclidean Poisson, whose coefficient is the reciprocal covolume $2/\sqrt{3}$, then has precisely the stated form. This is not an error in the source's Poisson formula: its measure is designed for lattice counting, not unitarity. □

3 Private transform and energy interfaces

The actual private proof passages read were TR1674, lines 95–113 and 168–360, and the volume 2 master-atoms chapter, lines 447–541 [5, 6]. This is 307 distinct lines, not a whole-corpus read. The former explicitly separates a finite positive energy identity from the arithmetic correlations needed to estimate it. The latter fixes the Mellin measure and logarithmic-coordinate normalization. We use those interfaces, not any claimed transfer of a Liouville or zeta theorem to cubic characters.

For $c \in \mathbb{R}$, let $g_c(t) = e^{ct} f(e^t)$. If $g_c \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, then

$$\mathcal{M}f(c+i\tau) = \int_0^\infty f(x) x^{c+i\tau} \frac{dx}{x}, \quad \int_0^\infty |f(x)|^2 x^{2c-1}\, dx = \frac{1}{2\pi} \int_{\mathbb{R}} |\mathcal{M}f(c+i\tau)|^2\, d\tau. \tag{3}$$

The proof is the substitution $x = e^t$ followed by ordinary Fourier Plancherel with frequency $-\tau$; inversion is in weighted L^2 unless stronger hypotheses give pointwise convergence. Dilation $f(bx)$ multiplies the transform by b^{-s}, $b > 0$. This does not justify a contour shift: poles, gamma growth, tails and conductor-dependent phases remain obligations. In the source smooth duality argument the root number, conductor phase and unit-modulus gamma quotient are row scalars on the symmetry line. Moving them into common coefficients would destroy the asserted common-polynomial structure, even though their absolute values equal one.

3.1 Complex corrected energy, with the phase retained

For finite sets of primary a, b, nonnegative weights w_a, and complex β_b, put

$$F(a) = \sum_b \beta_b G(b) \overline{\chi_b(a)}, \quad S = \sum_b \beta_b (\mathrm{N}\, b)^{-1/6}, \quad v(a) = c_* \overline{G(a)} (\mathrm{N}\, a)^{-1/6}.$$

Let $T_{ab} = G(b)\overline{\chi_b(a)}$, $q_b = (\mathrm{N}\, b)^{-1/6}$ and $D_w = \mathrm{diag}(w_a)$. Then exactly

$$\sum_a w_a |F(a) - v(a)S|^2 = \beta^*(T - vq^t)^* D_w (T - vq^t)\beta = \|F\|_w^2 - 2\,\mathrm{Re}(\overline{S}\langle F, v\rangle_w) + |S|^2 \|v\|_w^2, \tag{4}$$

where $\langle F, v\rangle_w = \sum_a w_a F(a)\overline{v(a)}$ and $\|F\|_w^2 = \sum_a w_a |F(a)|^2$ (similarly for v). This follows by expansion, with conjugate transpose, not transpose. The arithmetic mixed term contains $G(a)F(a)$, as in the source. A unitary change of the weighted output coordinates preserves this identity; a non-tight frame in general supplies only its frame bounds. Positivity of this Gram matrix alone supplies no saving, no SW estimate and no prime bias. The source correctly keeps the mixed term and cube-model square. We do not claim that a missing complex conjugation in that source has been repaired.

For comparison, the private prefix-energy identity has the transparent complex version: for $L_n = \sum_{j \le n} z_j$, $L_0 = 0$ and $N \ge 1$,

$$\sum_{n=1}^N \frac{|L_n|^2}{n(n+1)} = \sum_{n=1}^N \frac{|z_n|^2 + 2\,\mathrm{Re}(z_n \overline{L_{n-1}})}{n} - \frac{|L_N|^2}{N+1}.$$

Telescoping $1/[n(n+1)] = 1/n - 1/(n+1)$ proves it. The nonnegative left side does not bound the sign-sensitive correlations on the right. This is an explanatory classical identity, not a scored new theorem.

4 Full Poisson expansion of the cube contribution

The source's `dispersion.tex`, lines 499–591, parametrizes the cube frequencies, for squarefree primary d, by $h = d^2 j^3$. Each nonzero frequency has three preimages. Its fixed additive phase is one, and radiality removes a rotation. With

$$R^6 = \frac{27\,\mathrm{N}(b_1 b_2)}{A}, \qquad R \asymp H^{1/6}, \qquad H = \frac{B^2}{A}, \quad \mathcal{Q} = ABH^{1/3}, \tag{5}$$

the initially unrestricted cube sum is $\frac{1}{3}\sum_{j\in\mathcal{O}\setminus\{0\}} \widehat{W}((j/R)^3)$. The source replaces it by its area integral with error $O_W(R+1)$. Here is a stronger local replacement.

Theorem 4.1 (Explicit origin and rapid cube remainder)**.** *For $W \in \mathcal{S}(\mathbb{C})$, $\theta \in \mathbb{R}$, define $F_\theta(z) = \widehat{W}(e^{i\theta} z^3)$ and $C_{W,\theta}(R) = \frac{1}{3}\sum_{j\in\mathcal{O}\setminus\{0\}} F_\theta(j/R)$. For $R \geq 1$ and each integer $r \geq 2$,*

$$C_{W,\theta}(R) = \frac{R^2}{3} I_W - \frac{\widehat{W}(0)}{3} + E_{r,\theta}(R), \tag{6}$$

$$|E_{r,\theta}(R)| \leq \frac{R^{2-2r}}{3(16\pi^2)^r} \left\|\Delta^r F_\theta\right\|_{L^1(dm)} \sum_{k\in\mathcal{O}^*\setminus\{0\}} |k|^{-2r}, \tag{7}$$

$$I_W = c_*^2 \int_{\mathbb{C}} W(x)|x|^{-2/3}\, dm(x). \tag{8}$$

The constants are uniform in θ for fixed W. No radiality is needed for this isolated lattice statement.

Proof. The polynomial composition F_θ is Schwartz: each derivative is a finite sum of polynomial factors times derivatives of $\widehat{W}$ evaluated at a point of modulus $|z|^3$. Applying Poisson to $F_\theta(z/R)$ gives

$$\sum_{j\in\mathcal{O}} F_\theta(j/R) = R^2 \sum_{k\in\mathcal{O}^*} \mathcal{F}_m F_\theta(Rk).$$

The zero terms on the two sides are respectively $\widehat{W}(0)$ and $R^2 \int F_\theta\, dm$. Integration by parts gives $\mathcal{F}_m(\Delta^r F)(y) = (-16\pi^2|y|^2)^r \mathcal{F}_m F(y)$. Taking the L^1 bound proves (7); the lattice sum converges for $r > 1$. Rotation in z shows $\int F_\theta\, dm = \int F_0\, dm$ and $\|\Delta^r F_\theta\|_1 = \|\Delta^r F_0\|_1$.

The map $z \mapsto z^3$ has three sheets and real Jacobian $9|z|^4$, so

$$\int_{\mathbb{C}} \widehat{W}(z^3)\, dm(z) = \frac{1}{3}\int_{\mathbb{C}} \widehat{W}(y)|y|^{-4/3}\, dm(y).$$

For completeness, insert $|y|^{-4/3} = \Gamma(2/3)^{-1}\int_0^\infty t^{-1/3} e^{-t|y|^2}\, dt$. On compact t intervals Fubini and the two-dimensional Gaussian transform give the inner Euclidean integral $\pi t^{-1} e^{-4\pi^2|x|^2/t}$. Letting the interval expand yields

$$\int \widehat{W}(y)|y|^{-4/3}\, dm(y) = \frac{2}{\sqrt{3}}\, 2^{-2/3}\pi^{1/3}\frac{\Gamma(1/3)}{\Gamma(2/3)} \int W(x)|x|^{-2/3}\, dm(x).$$

Limits are justified by the integrable power singularities at zero and Schwartz decay; on the Gaussian side integrate the positive kernel against $|W|$. Dividing by three and using $\Gamma(1/3)\Gamma(2/3) = 2\pi/\sqrt{3}$ gives exactly c_*^2. Thus the source's Riesz constant is retained, not replaced by a unitary Fourier constant. □

Corollary 4.2 (Source-relative gain and the residual obstruction)**.** *For a fixed source weight and $H \geq 1$, the unrestricted cube main term has error $O_W(1)$ if the origin term is absorbed, instead of $O_W(H^{1/6}+1)$. Relative to its natural $H^{1/3}$ scale these are respectively $O_W(H^{-1/3})$ and $O_W(H^{-1/6})$. If the origin term is retained explicitly, the remaining relative error is $O_{W,r}(H^{-r/3})$ for every integer $r \geq 2$.*

Proof. Use $R \asymp H^{1/6}$ in (6)–(7). For nonnegative nonzero W, $\widehat{W}(0) = \int W\, dm > 0$, so the difference between the lattice sum and the area term tends to $-\widehat{W}(0)/3$. Without this secondary term a rapidly vanishing absolute error is generally false. □

The exact secondary coefficient in the source's $f = l = m = 1$ branch, before its remaining d, b_1, b_2 sums, is

$$-\frac{A\widehat{W}(0)}{27(\mathrm{N}\, d)^2\sqrt{\mathrm{N}(b_1 b_2)}}\beta_{b_1}(u)\overline{\beta_{b_2}(u)}. \tag{9}$$

Indeed multiply the omitted-origin term by the source prefactor $A/[9(\mathrm{N}\, d)^2|b_1 b_2|]$. The off-diagonal coprimality and sieve weights remain present; (9) is not automatically a signed rank-one global correction. For divisor-bounded coefficients, summing its absolute value is $O(AB(\log X)^C) = O(\mathcal{Q}H^{-1/3}(\log X)^C)$.

Lemma 4.3 (The retained coprimality cost)**.** *For a fixed Schwartz F, $R \geq 1$, and a primary squarefree b, removing the nonzero j divisible by some $\pi \mid b$ costs at most*

$$\sum_{\pi \mid b}\sum_{j \in \pi\mathcal{O}\setminus\{0\}} |F(j/R)| \ll_F R^2 \sum_{\pi \mid b}(\mathrm{N}\,\pi)^{-1}.$$

Proof. Put $t = R/|\pi|$. The lattice-point shell bound and Schwartz decay show

$$\sum_{k \neq 0} |F(e^{i\arg\pi}k/t)| \ll_F t^2 \qquad (t \geq 1).$$

For $t < 1$, a decay exponent $M > 2$ gives $O_F(t^M) \leq O_F(t^2)$. The nonzero condition is essential in the latter estimate. Sum over π; overcounting intersections only increases this upper bound. □

Thus roughness costs, the exact arithmetic diagonal, noncube frequencies and the mixed Type I term do not disappear. With $H \geq (\log X)^{C_g}$, the isolated area-only discretization condition changes from approximately $C_g/6 > K + C$ to $C_g/3 > K + C$. No improvement to the full permissible C_g, global logarithmic saving or first-moment exponent is asserted. If W varies with X, its displayed Laplacian seminorm must be charged; arbitrary power-width families do not inherit a free uniform rapid error. Nonradial W is allowed locally in the theorem, not automatically in the source's global Poisson/character separation.

5 Distance-weighted Perron cells

The source `perron.tex`, lines 210–312, divides both fixed log-norm intervals into $O(J)$ cells of width $1/J$. Write $a_i = \|\alpha 1_i\|_2$, $b_j = \|\beta 1_j\|_2$. Its arithmetic input gives, uniformly after every real common Mellin shift, the cell bound

$$Ca_i\left(\frac{A}{J}b_j^2 + \mathcal{Q}\varepsilon_J\right)^{1/2}. \tag{10}$$

Here $\varepsilon_J = (\log X)^{-K}$ for $J = (\log X)^{D'}$, or $\varepsilon_J = X^{-10\gamma}$ for $J = X^{\gamma}$. The source sums all absolute errors over all pairs, giving $J^{3/2}\sqrt{\varepsilon_J}$ in normalized units. The following conditional theorem retains their distance decay. It does *not* assume an error proportional to b_j.

Theorem 5.1 (Distance-sensitive cell summation)**.** *Fix an integer $q \geq 2$. Consider finite coefficients on the two dyads and the height integral*

$$\mathcal{I} = \int h_T(t) \sum_{a,b} \alpha_a \beta_b K(a,b) V(\mathrm{N}(ab)/X) e^{it \log(\mathrm{N}(ab)/X_0)}\, dt,$$

where $X_0 \asymp X$, $V \in C_c^\infty((0,\infty))$ is fixed, $h_T \in C_c^\infty(\mathbb{R})$ is supported on $T \leq |t| \leq 2T$, and $\left\|h_T^{(s)}\right\|_\infty \leq C_s T^{-s-1}$. Suppose (10) bounds the absolute value of this height integral restricted to each cell pair, after every real common Mellin shift $(\mathrm{N}\, a)^{iv}(\mathrm{N}\, b)^{iv}$ in the coefficients, and for all normalized height cutoffs obtained from h_T by at most q derivatives, with one constant. This is an integrated estimate, not a pointwise bound on the bare polynomial. Suppose there are at most $C_0 J$ cells per side. Then for $T \geq J \geq 1$,

$$|\mathcal{I}| \ll_{q,V,C_s,C_0} \sqrt{A/J}\, \|\alpha\|_2\, \|\beta\|_2 + \sqrt{\mathcal{Q}\varepsilon_J J}\, \|\alpha\|_2 . \tag{11}$$

For general $T > 0$, multiply this right side by $O_q(1 + (J/T)^q)$. No assertion about (10) for a new arithmetic kernel K is part of this theorem.

Proof. Write $s = J \log(\mathrm{N}(ab)/X_0)$. For a fixed pair of cells, s lies in an interval of uniformly bounded length with center $k = i + j - \sigma$, where σ is real and independent of i, j. Choose a fixed large c_0. If $|k| \leq c_0$, use (10). There are only boundedly many such pairs in each row and column.

For $|k| > c_0$, integrate by parts q times in t. Compact support removes every boundary term. The new height function is $T^q h_T^{(q)}$ up to a scalar of modulus one, the outside factor is $(J/T)^q$, and the product multiplier is s^{-q}. Let χ be smooth, compactly supported, and equal to one on an interval containing all the possible values of $s - k$ in this cell pair. Put $F_k(s) = \chi(s-k)s^{-q}$. With c_0 larger than the support radius,

$$\|F_k\|_1 + \|F_k''\|_1 \ll_q (1+|k|)^{-q}, \qquad \left\|\widehat{F_k}\right\|_1 \ll_q (1+|k|)^{-q}.$$

The second estimate follows by bounding the Fourier transform by $\|F_k\|_1$ for $|\xi| \leq 1$ and by $(2\pi|\xi|)^{-2}\|F_k''\|_1$ elsewhere. Scaling $s = J\ell$ does not change the Fourier L^1 norm. Its inverse Fourier representation therefore separates this multiplier into common real norm shifts, all independent of t. Multiplication by the fixed envelope convolves its integrable Mellin transform and changes only the constant. The unrestricted-shift hypothesis now applies. The resulting pair bound is

$$C w_{ij} a_i (\sqrt{A/J}\, b_j + \sqrt{\mathcal{Q}\varepsilon_J}), \quad w_{ij} = \begin{cases} 1, & |i+j-\sigma| \leq c_0, \\ (J/T)^q (1+|i+j-\sigma|)^{-q}, & \text{otherwise.} \end{cases}$$

Uniformly in real σ, row and column sums of w are bounded by $C_q(1 + (J/T)^q)$, since $q > 1$. The weighted Cauchy inequality gives

$$\sum_{i,j} w_{ij} a_i b_j \leq \left(\sum_{i,j} w_{ij} a_i^2\right)^{1/2} \left(\sum_{i,j} w_{ij} b_j^2\right)^{1/2} \ll_q (1 + (J/T)^q)\, \|\alpha\|_2\, \|\beta\|_2 .$$

For the absolute error, $\sum_{i,j} w_{ij} a_i \ll_q (1 + (J/T)^q) \sum_i a_i \ll_q (1 + (J/T)^q)\sqrt{J}\, \|\alpha\|_2$. These estimates prove the assertion, including empty or highly unequal-mass cells. The proof uses no cancellation between different cell pairs. □

Corollary 5.2 (Source parameter consequences)**.** *Under the same arithmetic input and source coefficient moment bounds,* (11) *is*

$$\ll X^{5/6}(\log X)^M\big(H^{-1/6}J^{-1/2}+J^{1/2}\varepsilon_J^{1/2}\big),\qquad T\ge J. \tag{12}$$

For the log-width regime with $H\gg 1$, sufficient local choices are $D'>2(D+M+2)$ and $K>D'+2(D+M+2)$, replacing the source condition $K>3D'+2(D+M+2)$ for these cell errors. For $J=X^{\gamma}$, the absolute-error factor is $X^{-9\gamma/2}$ instead of $X^{-7\gamma/2}$; the diagonal factor remains at most $X^{3\kappa/2-\gamma/2}$ when $H\gg X^{-9\kappa}$.

Proof. Use $\|\alpha\|_2\ll\sqrt{A}(\log X)^M$ and $\|\beta\|_2\ll\sqrt{B}(\log X)^M$, absorbing fixed powers into M. The identities $A\sqrt{B}=X^{5/6}H^{-1/6}$ and $\sqrt{AQ}=X^{5/6}$ hold when $AB=X$; comparability suffices otherwise. Substitution gives every exponent stated. □

The height threshold still must satisfy the source's averaged dispersion requirements and the undecomposed prime-model bound, as well as $T\ge J$. We do not obtain an explicit new global C_t, nor remove the small-B restriction. For $B>X^{.40}$ the common-shift uniformity used here is not available in this unrestricted form. Fixed angular twists stay in the coefficients and do not affect this conditional proof; its input constants may still depend on the fixed angular mode.

6 Sharp local margin in the Gram contradiction

In `dual.tex`, lines 400–482, the source first obtains, along a subsequence, a fixed positive short exponent $a\le 1/2$ whose value exponent equals $5a/6$. Repeating this factor m times gives length $Z=Y^{z+o(1)}$, $z=ma$, and values $Y^{5z/6+o(1)}$ on $Y^{2/3-o(1)}$ rows. After dividing the Gram inequality by ZR, the claimed left exponent and the diagonal and off-diagonal exponents are

$$E_L=\frac{2z}{3},\qquad E_D=z-\frac{2}{3},\qquad E_O=\frac{z+\max(1,2-2z/3)-2/3}{2}.$$

The row count, character Gram bound and arithmetic construction of the polynomial are inputs to the following elementary optimization, not its conclusions.

Proposition 6.1 (Optimal universal single-factor margin)**.** *For $4/3<z<2$ put*

$$g(z)=\min\left\{\frac{2-z}{3},\begin{cases}(3z-4)/6, & z\le 3/2,\\ (z-1)/6, & z\ge 3/2.\end{cases}\right\}.$$

Its maximum over real z is $1/9$, attained at $z=5/3$. For every fixed $0<a\le 1/2$ there exists an integer $m\ge 1$ with $g(ma)\ge 2/51$. The universal constant $2/51$ is sharp if the only choice allowed is an integer multiple of that one exponent a in $(4/3,2)$. One valid explicit rule is

$$m(a)=\begin{cases}\lceil 24/(17a)\rceil, & 0<a\le 8/17,\\ 3, & 8/17<a\le 1/2.\end{cases} \tag{13}$$

Proof. Subtract E_D,E_O from E_L to obtain the displayed two gaps. On $(4/3,3/2]$ the off-diagonal gap is the smaller and increases to $1/12$. On $[3/2,2)$ the two affine functions cross at $5/3$, where both are $1/9$; their minimum increases before that crossing and decreases after it.

Every $z\in[24/17,32/17]$ has diagonal gap at least $2/51$. If $z\le 3/2$, its other gap is at least $(3(24/17)-4)/6=2/51$. If $z\ge 3/2$, its other gap is at least $1/12$. When $a\le 8/17$, rule (13) places ma in $[24/17,24/17+a)\subset[24/17,32/17]$. When $8/17<a\le 1/2$, $3a\in(24/17,3/2]$. This proves the bound. At $a=8/17$, the only admissible multiples in $(4/3,2)$ are $3a=24/17$ and $4a=32/17$; both have margin $2/51$. Hence no larger uniform constant can hold for this strategy. □

The parent suggested the sharp interval and rule; the proof and executable checks here were independently worked through by this report's author. This is a sharpened local exponent certificate, not a new character Gram inequality. Its use in the source contradiction still requires all selected polynomial products to satisfy the source sieve hypotheses. In particular $m(a)$ is unbounded as $a \downarrow 0$. Divisor powers, support ratios, constants and convergence of limiting exponents can depend on this fixed multiplicity. The proposition therefore does not convert the source's compactness argument into an explicit uniform global η, nor improve its global first-moment exponent. If additionally $a \geq a_0 > 0$, it gives the finite bound $m \leq \max(3, \lceil 24/(17a_0) \rceil)$, but quantitative arithmetic losses and convergence rates still have to be supplied.

7 Comparison, obligations and feasible extensions

7.1 What is improved, and what is retained

The cube result replaces a first-order area comparison by a full Schwartz Poisson expansion, identifies a necessary secondary constant, and proves a uniform rotation version of that local statement. Its advantage is a strictly sharper, seminorm-controlled discretization error. Its limitation is that the origin and arithmetic diagonal both live at an AB-type scale, and roughness still controls the coprimality remainder.

The cell result reduces the needed absolute-error budget by a factor J and keeps the near-cell diagonal scale for the far cells as well. It works with arbitrarily uneven cell masses. Its limitation is the strong, explicit uniformity in common real shifts and normalized cutoff derivatives; without that input one cannot integrate the separated multipliers against the same estimate. It is a quantitative deduction from the source dispersion statement, not a proof of that statement.

The Gram result gives an optimal universal constant for a specified discrete choice problem. Its limitation is unbounded multiplicity at small a. None of these improvements follows merely by renaming a finite Gram matrix or changing coordinates. Classical Fourier, Mellin and positivity identities are preparatory tools, not credited as novel discoveries.

7.2 Source details that must not be lost

The source correctly preserves the exact diagonal in its height average, uses a squarefree majorant only in the positive square, and retains the true squarefree weight in the mixed and model terms. Its cancellation has coefficients $1, -2, 1$ with the complex real part in (4). It keeps conductor phases as row scalars in duality, uses independent smooth prime weights in the large-B range, and separates stopped coefficients before estimating either kernel. Those details are not optional.

Its angular proof does not infer twisted SW from SW for arbitrary untwisted sequences; it checks the actual prime-convolution coefficients. The mode is fixed, and the identity $G(\pi)^3 = -\pi/|\pi|$ gives the stated nontrivial fixed-power consequences only for powers not divisible by three. The source itself does not obtain the stronger $X^{5/6}/\log X$ cancellation for nonzero multiple-of-three powers from its Hecke prime estimate. We retain this limit.

The public normalization conflict is the substantive compatibility warning in this report. The lattice transform factor $1/3$, the Riesz constant c_*^2, and the complex corrected-square signs are not identified as source errors. The sharper cube error and cell budget replace valid but weaker bounds. The source's unconditional global claim remains a source claim, particularly while its compatibility with the cited rational-prime theorem is unresolved.

7.3 Named unresolved inputs

For clarity, the following are not supplied by the local propositions: the cubic large sieve and the claimed structured exceptional-moment saving; the level Voronoi coefficient/residue formulas and ramified indexing; the metaplectic height mean square and conductor growth estimates; uniform SW for the actual rough sequences with exclusions; the entire low/high-height assembly and Perron error at the final scale. The source's Type I derivation reads a theta index from the proof formula (5.79) of its cited reference rather than its printed (5.67); this cross-reference is retained as an imported normalization issue, not independently rederived from the automorphic construction here.

Heath-Brown's primary repository record describes an unconditional upper bound, whereas Dunn–Radziwiłł state their prime asymptotic under GRH [4, 3]. The present search and scoped reading do not establish that no other work exists, historical priority for our local deductions, or that an unconditional asymptotic follows from a finite test. The dated OpenAI preprint is a current source claim, not certification by provenance.

7.4 Next mathematical work

A focused next step is a coefficient-level reconciliation of the rational smoothed and all-primary normalizations, through both the exact conjugate pairing and the metaplectic residue. A separate task is a complete quantitative ledger for the exceptional moment, including multiplicity-dependent divisor constants; the uniform 2/51 local margin alone is insufficient. For the present improvements, one can optimize the cube cutoff only after combining its origin, diagonal and roughness terms, or replace the one-dimensional cell distance by a proved summable interaction kernel in a genuinely higher-rank localization. Neither extension is asserted without its corresponding arithmetic input and measure normalization.

8 Verification and evidence boundaries

The accompanying author-run script passed 3769 bounded exact checks. It uses exact rational arithmetic for finite complex-energy and weighted-cell checks, and exact integer polynomial reduction in cyclotomic rings for small prime Gauss sums. It verifies the trace lattice determinant and dual basis, the sharp/smooth coefficient conversion, the Gram gaps and the explicit multiplicity rule. A separate floating-point Gaussian cube-lattice experiment illustrates the origin term; it is labelled numerical, not an exact infinite-lattice certificate. There are 448 complex-energy checks, 576 cell-kernel checks, 1801 Gram exponent checks, 900 finite-field/cyclotomic checks, and 44 normalization or error-exponent checks. Four Gaussian experiments use $R = 1, 2, 4, 8$; at $R = 8$ the area-only residual is approximately -0.3849003651, while subtracting the predicted origin term leaves approximately $-1.86 \cdot 10^{-7}$. Those four numerical experiments are not in the exact count. Test counts and parameters are in `M049-checks.json`. Parent checks are not included.

All eight public TeX files were literally read, but external large-sieve, Hecke and metaplectic proofs were not fully reread. Private scopes comprise only the 307 lines specified above. The completed read ledger records all unread complements and matching pre/post hashes, which establish endpoint byte identity, not continuous monitoring or independent authentication of the repository history. The parent handles that separate provenance task. The final build record distinguishes the native compiler attempt from the existing MiKTeX export fallback. Every final PDF page is rendered and visually inspected; the final freeze binds outputs and owned evidence, not mutable parent or global files.

The C3 assessment rests on Theorem 4.1, Theorem 5.1, Corollary 5.2 and Proposition 6.1. It is independent of L3 and does not assess the global claimed theorem as proved. The use of established analytic tools and parent-suggested exponent optimization is disclosed; no historical novelty, global power saving or resolution of Patterson's normalization is claimed.

References

[1] OpenAI, *An unconditional first moment for cubic Gauss sums*, September 25, 2026. Assigned cached build, eight TeX files. Pinned public source directory.

[2] A. Dunn and M. Radziwiłł, *Bias in cubic Gauss sums: Patterson's conjecture*, arXiv:2109.07463v3, May 14, 2024. Primary HTML, Introduction, Theorem 1.1 and equations (1.6)–(1.8). Only the displayed introduction and metadata were externally inspected here; the full proof remains imported.

[3] A. Dunn and M. Radziwiłł, same title, *Annals of Mathematics* 200 (2024), 967–1057. Journal abstract and publication record.

[4] D. R. Heath-Brown, *Kummer's conjecture for cubic Gauss sums*, *Israel Journal of Mathematics* 120 (2000), 97–124. Oxford primary repository record. Repository abstract/metadata inspected; full imported sieve and mean-square proofs not audited in this report.

[5] M. Subbarao, private technical report TR1674, revised v2. Local original proof passages, lines 95–113 and 168–360; exact path and SHA256 in the confidential local read ledger. No public upload.

[6] M. Subbarao, private *Ram Calculus*, volume 2, master-atoms chapter, lines 447–541: Mellin transform, log coordinates, inversion and weighted Plancherel. Exact local path and SHA256 in the ledger.